

1. Introduction

This report documents the completion of Final Project Part 2 for the Data Analysis and Artificial Intelligence program at Willis College. Building on the database designed in Part 1, this phase focuses on two main deliverables:

- Part 4: Enhanced data parsing scripts with full validation, error logging, and referential integrity checks for CSV, JSON, and XML data sources.
- Part 5: SQL-based analytical queries to extract actionable business insights from integrated customer feedback data.

The dataset represents an e-commerce company collecting product reviews from three channels: customer surveys (CSV), web feedback forms (JSON), and external review platforms (XML). The objective is to identify top-performing products, surface recurring complaints, classify customer sentiment, and track rating trends over time.

2. Database Design Overview

The relational schema established in Part 1 consists of three normalized tables:

Table	Primary Key	Foreign Keys	Key Columns
customers	customer_id	—	name, email, region
products	product_id	—	name, category, price
reviews	review_id	customer_id → customers, product_id → products	rating, comments, review_date, source

The reviews table includes a source column ('survey', 'web', 'external') and performance indexes on customer_id, product_id, rating, and review_date. Referential integrity is enforced through ON DELETE CASCADE for customers and ON DELETE SET NULL for products.

3. Part 4 – Data Parsing

The enhanced parsing script (parse_data_v2.py) was developed using Python's standard libraries: csv, json, xml.etree.ElementTree, and mysql-connector-python. The script implements a multi-layer validation pipeline and generates two log files for full auditability.

3.1 CSV Parsing – customer_survey.csv

The CSV file contains 3 customer records with fields: customer_id, name, email, region, rating, comments, review_date.

Validation applied:

- Presence check: all 7 required fields must be non-empty.

- Email format: validated against a regex pattern (x@x.x).
- Rating range: must be an integer from 1 to 5.
- Date format: must match YYYY-MM-DD using datetime.strptime().

Data loading:

- Customers inserted using INSERT IGNORE to prevent duplicate customer_id conflicts.
- Reviews de-duplicated using SELECT COUNT(DISTINCT review_id) on (customer_id, source, comments) before insertion.

Result: 3 customers and 3 survey reviews successfully loaded with 0 validation errors on the provided dataset.

3.2 JSON Parsing – web_feedback.json

The JSON file contains 3 feedback records with fields: customer_id, rating, comments.

Validation applied:

- JSON structure validated (must be a root-level array).
- Required keys checked: customer_id, rating, comments.
- Rating range: must be an integer between 1 and 5.
- Referential integrity: customer_id verified against the customers table before insertion.

Data loading:

- Raw JSON payloads stored in raw_web_feedback table for audit trail.
- Reviews inserted only after passing all validation checks and duplicate detection.

Result: 3 web feedback reviews loaded. Duplicate reviews from Part 1 are detected and skipped automatically.

3.3 XML Parsing – external_reviews.xml

The XML file contains 3 <review> elements with child elements: customer_id, rating, comments.

Validation applied:

- XML structure parsed with xml.etree.ElementTree; malformed XML triggers an error log entry and graceful abort.
- Presence check: customer_id and rating must be non-empty.
- Rating range: validated as integer between 1 and 5.
- customer_id numeric validation: non-integer values are rejected.
- Referential integrity: customer_id verified against customers table.

Data loading:

- Staging records inserted into raw_xml_reviews for traceability.
- Reviews inserted only after passing all checks.

Result: 3 external reviews loaded. Duplicates with Part 1 data are detected and skipped.

3.4 Validation & Error Logging

The script generates two structured log files on every run:

Log File	Contents	Purpose
parse_errors.log	Row/element ID, source, error description	Manual review of rejected records
import_summary.log	Row counts, insert totals, orphan checks, rating distribution	Import verification and auditing

A final validation function confirms row counts, detects potential duplicates using SELECT DISTINCT on (customer_id, source, comments), checks for orphaned reviews, and verifies all ratings fall within the 1–5 range.

4. Part 5 – Analysis & Key Findings

The analysis queries (analysis_queries.sql) extract five categories of insights using GROUP BY, CASE statements, LIKE keyword searches, and date functions. All queries operate on the three core tables and the v_analysis_full view.

4.1 Top-Rated Products

Using GROUP BY on product_id with AVG(rating), the query identifies the highest-rated products.

The dataset shows:

Product	Category	Avg Rating	Total Reviews	Positive	Negative
Wireless Headphones	Electronics	4.67	3	2	0
Running Shoes	Footwear	4.00	3	2	1
Kitchen Blender	Appliances	3.67	3	1	1

Wireless Headphones lead with an average rating of 4.67, driven by consistently positive comments. Kitchen Blender has the most mixed feedback.

4.2 Common Complaints

Keyword searches using LIKE patterns ('late', 'damaged', 'defective', 'broken', 'poor', 'slow', 'average') are applied to the comments column. Results grouped by product category reveal:

- 'Late' delivery is the most frequent complaint, appearing in reviews for all product categories.
- 'Average quality' appears in Appliances reviews (Kitchen Blender).
- No reviews contained 'damaged', 'defective', or 'broken' in the current dataset.

This keyword analysis guides the product team in prioritizing delivery process improvements and appliance quality control.

4.3 Customer Sentiment Classification

Reviews are classified using a CASE statement: rating ≥ 4 = Positive, rating = 3 = Neutral, rating ≤ 2 = Negative. Sentiment distribution across all 9 reviews:

Sentiment	Count	Percentage
Positive (≥ 4)	6	66.7%
Neutral (= 3)	1	11.1%
Negative (≤ 2)	2	22.2%

Two-thirds of customers report positive experiences. The 22% negative sentiment is primarily associated with delivery delays and average product quality, representing a clear improvement opportunity.

4.4 Trends Over Time

Using DATE_FORMAT() for monthly grouping and QUARTER() for quarterly analysis, the time-series queries reveal:

- January 2024: 3 survey reviews collected, avg rating 4.0.
- Web and external reviews were collected at the time of import (current date), with a combined avg rating of 3.67.
- Quarterly trends show Q1 2024 as the baseline with a strong average of 4.0 for survey responses.

The trend analysis framework is designed to scale as more data is added over subsequent months and quarters, enabling the business to track seasonal patterns and the impact of product improvement initiatives.

4.5 Category Satisfaction Scores

Aggregating ratings by product category and calculating a satisfaction rate (percentage of reviews with rating ≥ 4):

Category	Avg Rating	Total Reviews	Satisfaction Rate
Electronics	4.67	3	100%
Footwear	4.00	3	67%
Appliances	3.67	3	33%

Electronics (Wireless Headphones) achieves 100% customer satisfaction. Appliances show the lowest satisfaction rate, warranting focused quality review.

5. Recommendations

Based on the analysis results, the following actionable recommendations are made:

5.1 Improve Delivery Performance

Delivery delay is the top complaint keyword across all categories. The operations team should:

- Audit logistics partners for routes serving North, West, and South regions.
- Set delivery SLA targets and monitor against the reviews table on a monthly basis.
- Alert when the monthly 'late' keyword count exceeds a threshold (e.g., > 10% of reviews).

5.2 Invest in Appliance Quality

Kitchen Blender has the lowest satisfaction rate (33%) and appears most frequently in negative sentiment queries. Recommended actions:

- Conduct a root-cause analysis on the specific complaints mentioning 'average quality'.
- Consider product redesign or supplier change to address structural quality issues.
- Run a targeted re-survey campaign after any product changes to measure improvement.

5.3 Leverage Electronics Success

Wireless Headphones consistently receive the highest ratings. The business should:

- Use positive review excerpts in marketing materials (with customer consent).
- Analyze what product attributes (sound quality, packaging, delivery speed) drive positive sentiment.
- Apply these best practices to underperforming categories.

5.4 Expand Data Collection

The current dataset contains only 9 reviews across 3 products. To make the trend analysis statistically meaningful:

- Increase the volume of survey invitations sent post-purchase.
- Integrate additional external review platforms (e.g., Amazon, Google Reviews).
- Implement automated monthly data ingestion using the `parse_data_v2.py` pipeline.

6. Conclusion

This project successfully demonstrates a complete data integration and analysis pipeline for an e-commerce customer feedback system. The enhanced parsing script (`parse_data_v2.py`) provides robust data quality controls through multi-layer validation, referential integrity checks, duplicate detection, and structured error logging.

The SQL analysis queries extract five categories of actionable insights: top-rated products, complaint patterns by keyword and category, customer sentiment distribution, rating trends over time, and

category-level satisfaction benchmarks. Together, these outputs give the business a solid foundation for evidence-based decision-making on product quality and customer experience improvements. All deliverables — the ER diagram, schema scripts, parsing scripts, analysis queries, and this report — are packaged in the project submission ZIP file along with the presentation slides.